AGREEMENT OF THE TWO DERIVED UPPER SHRIEK FUNCTORS FOR QUASI-FINITE MORPHISMS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The derived upper shriek Rf^! as right adjoint to Rf_! for separated finite type morphisms of quasi-compact quasi-separated schemes and torsion coefficient ring, obtained via Brown representability (more-etale, Lemma lemma-upper-shriek-derived, canonical tag 0G2C). cited in statement at line 4079. Defines the first of the two functors claimed to agree.
- Adjoint derived functors f_! and Rf^! on derived categories induced by the lower and upper shriek for locally quasi-finite morphisms of schemes with arbitrary coefficient ring (more-etale, Lemma lemma-lqf-shriek-derived, canonical tag 0F5N). cited in statement at line 4080. Defines the second of the two functors claimed to agree; its left adjoint is f_!.
- Agreement of the compactification-constructed Rf_! with the restriction of the locally quasi-finite f_! on derived categories to their common domains, for quasi-finite separated morphisms of quasi-compact quasi-separated schemes (more-etale, Lemma lemma-Rf-shriek-for-quasi-finite, canonical tag 0GKM). cited at proof line 4085. The named input giving Rf_! = f_!, so the two Rf^! functors have the same left adjoint.

Source reference. Original lemma 0G2C: sources/stacks-chow/tags.txt, source lines 20052–20052. Cited in the statement at line 4079: defines the first functor Rf^! claimed to agree.

Source reference. Original lemma 0F5N: sources/stacks-chow/tags.txt, source lines 18943–18943. Cited in the statement at line 4080: defines the second functor Rf^! claimed to agree.

Source reference. Original lemma 0GKM: sources/stacks-chow/tags.txt, source lines 20692–20692. Cited at proof line 4085: gives Rf_! = f_!, the identity of left adjoints behind the uniqueness argument.

Lemma 0.2. *Let $f : X \to Y$ be a separated quasi-finite morphism of quasi-compact and quasi-separated schemes. Let Λ be a torsion coefficient ring. The functor*

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$Rf^! : D(Y_{étale}, \Lambda) \to D(X_{étale}, \Lambda)$ of Lemma 0G2C is the same as the functor $Rf^!$ of Lemma 0F5N.

Proof. Follows from uniqueness of adjoints as $Rf_! = f_!$ by Lemma 0GKM. □